GPU Use Case Documentation

Startup: Jami Studio
Founder: James
Today's Date: June 11, 2026

Company Overview

Company: Jami Studio
Contact: James, solo technical founder
Website: https://Jami Studio

Executive Summary

Jami Studio is an open-core agent infrastructure platform (governed runtime, UI registry, multi-agent orchestration, provenance-backed knowledge layer, and agent governance). We are not training foundation models. Our GPU needs are inference-shaped and batch-shaped: accelerating knowledge ingestion, running self-hosted reference inference for our BYOK stack, and hosting evaluation workloads as we scale the commercial lane.
Primary user-facing voice and avatar workloads run through API partners (ElevenLabs, Anam) today. GPUs on DigitalOcean are for backend infrastructure we own: embeddings at scale, optional local model inference, and agent eval pipelines.

1. Resource Requirements

Phase 1 (Months 0-3): Development and Proof Workloads

Resource	Quantity	Notes
Single-GPU Droplet (L40S, A100 or equiv.)	1 instance	Primary dev/staging for Intercal batch jobs and Harness inference eval
GPU Hours (Burst)	~200-400 hrs/mo	Batch ingestion windows and evaluation runs

Phase 2 (Months 4-12): Staging and Early Hosted Lane

Resource	Quantity	Notes
Single-GPU Droplet	1-2 instances	Second node for heavy Intercal backfills or parallel eval suites
Peak Concurrent GPUs	2 max	Right-sized for solo founder scale

Our workloads are batch embedding, inference serving for smaller open models, and evaluation. We are not requesting multi-node H100x8 training clusters; a single-GPU credit package matches our technical shape.

Preferred Hardware SKUs

1. NVIDIA L40S or A100-class single GPU (Optimal for inference + batch embeddings)
2. Single H100 80GB (Same workloads, higher throughput)
3. AMD MI300X/MI325X (Alternative for inference batches)

2. GPU Use Cases

A. Intercal Knowledge Ingestion

Intercal is our provenance-backed temporal knowledge substrate. It ingests sources, extracts claims, resolves entities, and stores embeddings alongside Postgres/pgvector.

Current Status: local CPU embeddings via fastembed (ONNX) for development. Works for small batches, does not scale cleanly when we run large backfills across many sources.

GPU Utilization

- Batch embedding generation during ingestion and re-indexing (millions of chunks)
- Re-embedding for model upgrades
- Entity resolution and clustering experiments

Why GPU: Embedding batches are parallel and memory-bound. A single inference GPU cuts Intercal backfill time from hours on CPU to minutes, which matters for a solo founder running real pipelines.
Expected load: Bursty. Heavy during ingestion sprints and model upgrades; light day-to-day.

B. Self-Hosted Inference Reference

The Jami Agent Harness supports BYOK and self-host deployments. Most users will use API providers, but we need a credible self-hosted reference for OSS adopters.

GPU Utilization

Running open-weight models (7B-70B) for inference testing
Validating latency and throughput on DigitalOcean infrastructure

C. Evaluation and Benchmarking

We run structured evals when swapping models, voice providers, and retrieval settings.

GPU Utilization

Batch eval jobs over fixed prompt/response datasets
Retrieval quality benchmarks for Intercal corpora

5. Usage Timeline

As we open the hosted agent lane, we may offer optional local inference for specific workloads (privacy-sensitive tenants, air-gapped style deployments, cost caps).

GPU Utilization

Staging inference endpoint behind our gateway/LLM port
Optional RAG retrieval acceleration co-located with embedding indexes
Long-running agent "kitchen" containers that may attach a GPU for local model calls on specific tool paths

Why GPU: Keeps sensitive inference on infrastructure we control without forcing every customer through external APIs.

Expected load: Low initially. Ramp only with paying hosted users.

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6. Exclusions

The following activities and services are outside the scope of our GPU requirements:

Primary voice synthesis: ElevenLabs API (startup grant applicant)
Interactive video avatars: Anam API
Foundation model training: Not on our roadmap
Large-scale multi-node training: Not applicable

Crypto/mining: N/A

Planned Duration of GPU Use

Period	Plan
Months 0-3	1 single-GPU Droplet for Intercal batch embedding and inference evaluation.
Months 4-6	Maintain footprint. Burst to second GPU only for parallel backfills if volume demands.
Months 7-12	1-2 single-GPU nodes for staging and early hosted-lane inference experiments.

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7. Technical Infrastructure

Software Stack: TypeScript/Next.js, Python pipelines, Neon Postgres, containerized services via CUDA-compatible stacks (vLLM, Ollama).

8. Summary Table for Reviewers

Item	Answer
Preferred GPUs	Single-GPU Inference (L40S / A100-class)
Quantity	1 baseline, up to 2 peak
Primary Use Cases	Batch embeddings, inference eval, benchmarking
Duration	12 months, bursty utilization
Training at scale	No

James
Founder, Jami Studio